

**ALGEBRA I**
(Common Core)

Tuesday, June 13, 2017 — 1:15 to 4:15 p.m., only

Student Name _____

School Name _____

The possession or use of any communications device is strictly prohibited when taking this examination. If you have or use any communications device, no matter how briefly, your examination will be invalidated and no score will be calculated for you.

Print your name and the name of your school on the lines above.

A separate answer sheet for **Part I** has been provided to you. Follow the instructions from the proctor for completing the student information on your answer sheet.

This examination has four parts, with a total of 37 questions. You must answer all questions in this examination. Record your answers to the Part I multiple-choice questions on the separate answer sheet. Write your answers to the questions in **Parts II, III, and IV** directly in this booklet. All work should be written in pen, except for graphs and drawings, which should be done in pencil. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale.

The formulas that you may need to answer some questions in this examination are found at the end of the examination. This sheet is perforated so you may remove it from this booklet.

Scrap paper is not permitted for any part of this examination, but you may use the blank spaces in this booklet as scrap paper. A perforated sheet of scrap graph paper is provided at the end of this booklet for any question for which graphing may be helpful but is not required. You may remove this sheet from this booklet. Any work done on this sheet of scrap graph paper will *not* be scored.

When you have completed the examination, you must sign the statement printed at the end of the answer sheet, indicating that you had no unlawful knowledge of the questions or answers prior to the examination and that you have neither given nor received assistance in answering any of the questions during the examination. Your answer sheet cannot be accepted if you fail to sign this declaration.

Notice ...

A graphing calculator and a straightedge (ruler) must be available for you to use while taking this examination.

DO NOT OPEN THIS EXAMINATION BOOKLET UNTIL THE SIGNAL IS GIVEN.

High School Math Reference Sheet

1 inch = 2.54 centimeters	1 kilometer = 0.62 mile	1 cup = 8 fluid ounces
1 meter = 39.37 inches	1 pound = 16 ounces	1 pint = 2 cups
1 mile = 5280 feet	1 pound = 0.454 kilogram	1 quart = 2 pints
1 mile = 1760 yards	1 kilogram = 2.2 pounds	1 gallon = 4 quarts
1 mile = 1.609 kilometers	1 ton = 2000 pounds	1 gallon = 3.785 liters
		1 liter = 0.264 gallon
		1 liter = 1000 cubic centimeters

Triangle	$A = \frac{1}{2}bh$
Parallelogram	$A = bh$
Circle	$A = \pi r^2$
Circle	$C = \pi d$ or $C = 2\pi r$
General Prisms	$V = Bh$
Cylinder	$V = \pi r^2 h$
Sphere	$V = \frac{4}{3}\pi r^3$
Cone	$V = \frac{1}{3}\pi r^2 h$
Pyramid	$V = \frac{1}{3}Bh$

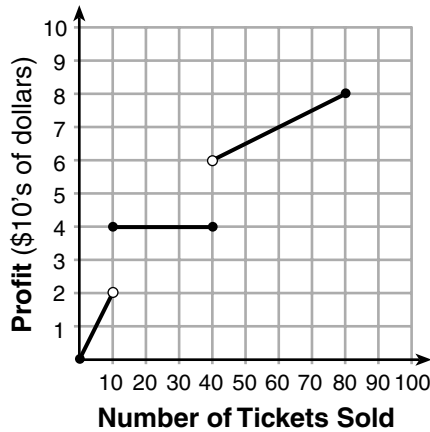
Pythagorean Theorem	$a^2 + b^2 = c^2$
Quadratic Formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Arithmetic Sequence	$a_n = a_1 + (n - 1)d$
Geometric Sequence	$a_n = a_1 r^{n-1}$
Geometric Series	$S_n = \frac{a_1 - a_1 r^n}{1 - r}$ where $r \neq 1$
Radians	1 radian = $\frac{180}{\pi}$ degrees
Degrees	1 degree = $\frac{\pi}{180}$ radians
Exponential Growth/Decay	$A = A_0 e^{k(t - t_0)} + B_0$

Part I

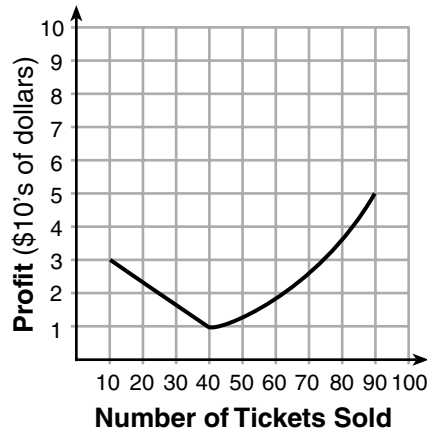
Answer all 24 questions in this part. Each correct answer will receive 2 credits. No partial credit will be allowed. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For each statement or question, choose the word or expression that, of those given, best completes the statement or answers the question. Record your answers on your separate answer sheet. [48]

Use this space for
computations.

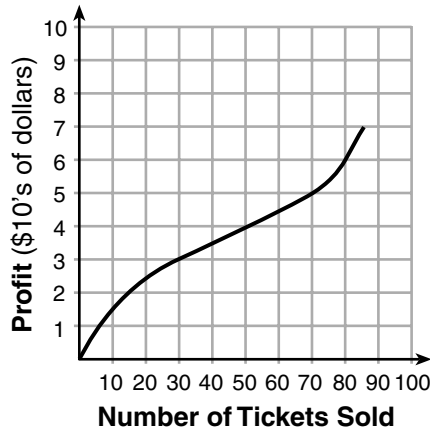
- 1 To keep track of his profits, the owner of a carnival booth decided to model his ticket sales on a graph. He found that his profits only declined when he sold between 10 and 40 tickets. Which graph could represent his profits?



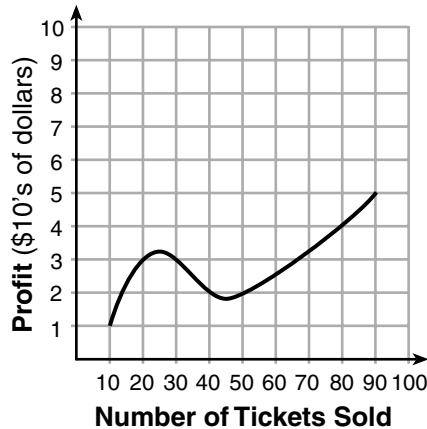
(1)



(3)



(2)



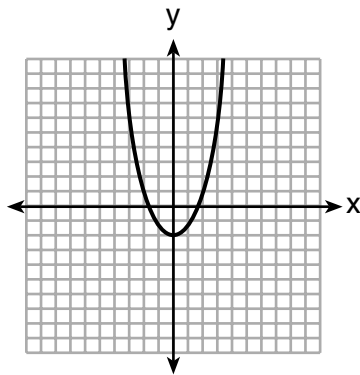
(4)

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computations.

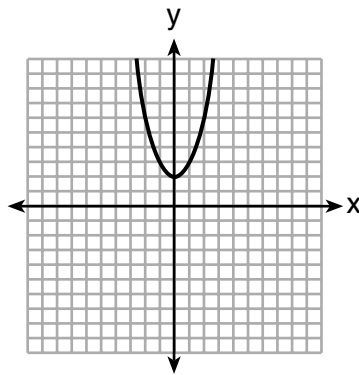
2 The formula for the surface area of a right rectangular prism is $A = 2lw + 2hw + 2lh$, where l , w , and h represent the length, width, and height, respectively. Which term of this formula is *not* dependent on the height?

- (1) A (3) $2hw$
(2) $2lw$ (4) $2lh$

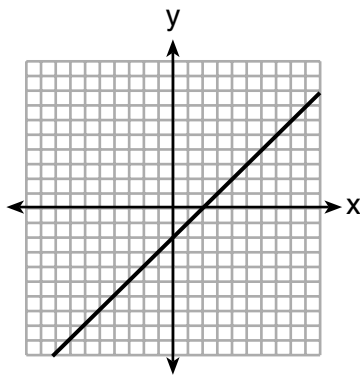
3 Which graph represents $y = \sqrt{x - 2}$?



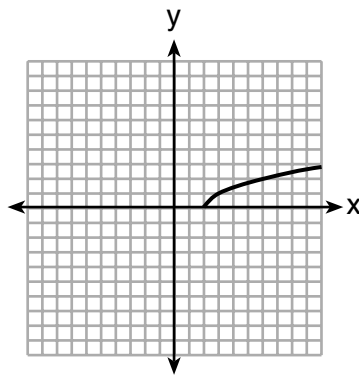
(1)



(3)



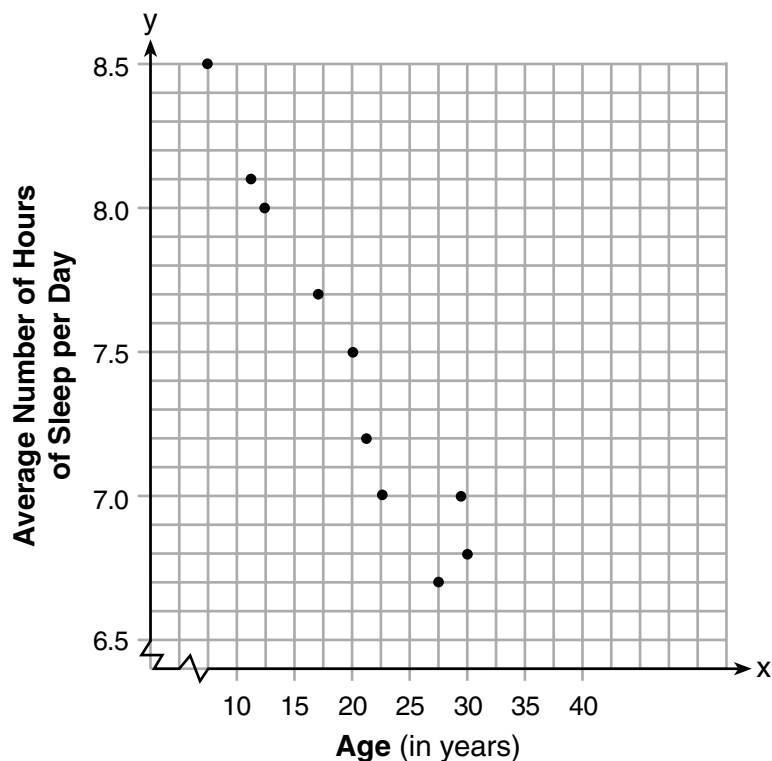
(2)



(4)

- 4 A student plotted the data from a sleep study as shown in the graph below.

Use this space for computations.



The student used the equation of the line $y = -0.09x + 9.24$ to model the data. What does the rate of change represent in terms of these data?

- (1) The average number of hours of sleep per day increases 0.09 hour per year of age.
- (2) The average number of hours of sleep per day decreases 0.09 hour per year of age.
- (3) The average number of hours of sleep per day increases 9.24 hours per year of age.
- (4) The average number of hours of sleep per day decreases 9.24 hours per year of age.

Use this space for
computations.

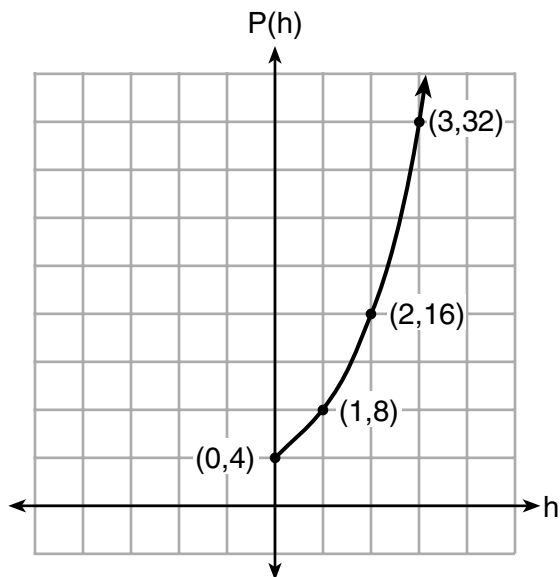
- 5 Lynn, Jude, and Anne were given the function $f(x) = -2x^2 + 32$, and they were asked to find $f(3)$. Lynn's answer was 14, Jude's answer was 4, and Anne's answer was ± 4 . Who is correct?

- (1) Lynn, only (3) Anne, only
(2) Jude, only (4) Both Lynn and Jude

- 6 Which expression is equivalent to $16x^4 - 64$?

- (1) $(4x^2 - 8)^2$ (3) $(4x^2 + 8)(4x^2 - 8)$
(2) $(8x^2 - 32)^2$ (4) $(8x^2 + 32)(8x^2 - 32)$

- 7 Vinny collects population data, $P(h)$, about a specific strain of bacteria over time in hours, h , as shown in the graph below.



Which equation represents the graph of $P(h)$?

- (1) $P(h) = 4(2)^h$ (3) $P(h) = 3h^2 + 0.2h + 4.2$
(2) $P(h) = \frac{46}{5}h + \frac{6}{5}$ (4) $P(h) = \frac{2}{3}h^3 - h^2 + 3h + 4$

**Use this space for
computations.**

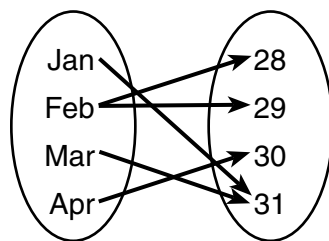
8 What is the solution to the system of equations below?

$$y = 2x + 8$$

$$3(-2x + y) = 12$$

- (1) no solution (3) $(-1, 6)$
(2) infinite solutions (4) $(\frac{1}{2}, 9)$

9 A mapping is shown in the diagram below.



This mapping is

- (1) a function, because Feb has two outputs, 28 and 29
(2) a function, because two inputs, Jan and Mar, result in the output 31
(3) not a function, because Feb has two outputs, 28 and 29
(4) not a function, because two inputs, Jan and Mar, result in the output 31

10 Which polynomial function has zeros at -3 , 0 , and 4 ?

- (1) $f(x) = (x + 3)(x^2 + 4)$ (3) $f(x) = x(x + 3)(x - 4)$
(2) $f(x) = (x^2 - 3)(x - 4)$ (4) $f(x) = x(x - 3)(x + 4)$

- 11** Jordan works for a landscape company during his summer vacation. He is paid \$12 per hour for mowing lawns and \$14 per hour for planting gardens. He can work a maximum of 40 hours per week, and would like to earn at least \$250 this week. If m represents the number of hours mowing lawns and g represents the number of hours planting gardens, which system of inequalities could be used to represent the given conditions?

- | | |
|---|---|
| (1) $m + g \leq 40$
$12m + 14g \geq 250$ | (3) $m + g \leq 40$
$12m + 14g \leq 250$ |
| (2) $m + g \geq 40$
$12m + 14g \leq 250$ | (4) $m + g \geq 40$
$12m + 14g \geq 250$ |

- 12** Anne invested \$1000 in an account with a 1.3% annual interest rate. She made no deposits or withdrawals on the account for 2 years. If interest was compounded annually, which equation represents the balance in the account after the 2 years?

- | | |
|-----------------------------|---------------------------|
| (1) $A = 1000(1 - 0.013)^2$ | (3) $A = 1000(1 - 1.3)^2$ |
| (2) $A = 1000(1 + 0.013)^2$ | (4) $A = 1000(1 + 1.3)^2$ |

- 13** Which value would be a solution for x in the inequality $47 - 4x < 7$?

- | | |
|-----------|----------|
| (1) -13 | (3) 10 |
| (2) -10 | (4) 11 |

- 14** Bella recorded data and used her graphing calculator to find the equation for the line of best fit. She then used the correlation coefficient to determine the strength of the linear fit.

Which correlation coefficient represents the strongest linear relationship?

- | | |
|-----------|------------|
| (1) 0.9 | (3) -0.3 |
| (2) 0.5 | (4) -0.8 |

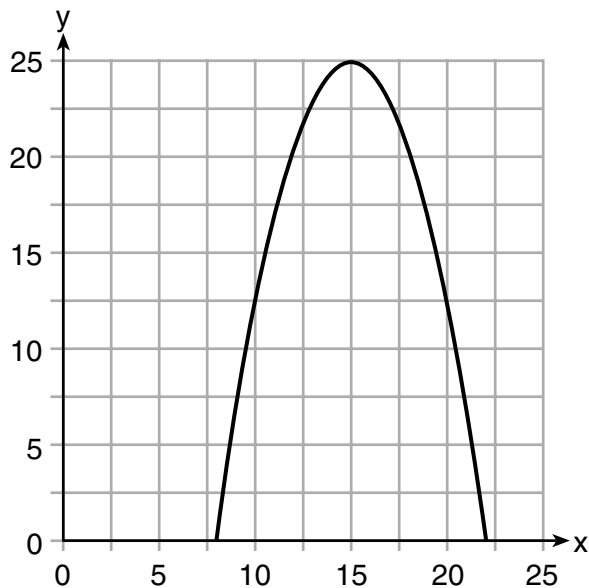
15 The heights, in inches, of 12 students are listed below.

61, 67, 72, 62, 65, 59, 60, 79, 60, 61, 64, 63

Which statement best describes the spread of these data?

- (1) The set of data is evenly spread.
- (2) The median of the data is 59.5.
- (3) The set of data is skewed because 59 is the only value below 60.
- (4) 79 is an outlier, which would affect the standard deviation of these data.

16 The graph of a quadratic function is shown below.



An equation that represents the function could be

- (1) $q(x) = \frac{1}{2}(x + 15)^2 - 25$
- (2) $q(x) = -\frac{1}{2}(x + 15)^2 - 25$
- (3) $q(x) = \frac{1}{2}(x - 15)^2 + 25$
- (4) $q(x) = -\frac{1}{2}(x - 15)^2 + 25$

Use this space for
computations.

- 17 Which statement is true about the quadratic functions $g(x)$, shown in the table below, and $f(x) = (x - 3)^2 + 2$?

x	$g(x)$
0	4
1	-1
2	-4
3	-5
4	-4
5	-1
6	4

- (1) They have the same vertex.
- (2) They have the same zeros.
- (3) They have the same axis of symmetry.
- (4) They intersect at two points.

- 18 Given the function $f(n)$ defined by the following:

$$f(1) = 2$$
$$f(n) = -5f(n - 1) + 2$$

Which set could represent the range of the function?

- (1) $\{2, 4, 6, 8, \dots\}$
- (2) $\{2, -8, 42, -208, \dots\}$
- (3) $\{-8, -42, -208, 1042, \dots\}$
- (4) $\{-10, 50, -250, 1250, \dots\}$

- 19 An equation is given below.

$$4(x - 7) = 0.3(x + 2) + 2.11$$

The solution to the equation is

- (1) 8.3
- (2) 8.7
- (3) 3
- (4) -3

**Use this space for
computations.**

- 20** A construction worker needs to move 120 ft^3 of dirt by using a wheelbarrow. One wheelbarrow load holds 8 ft^3 of dirt and each load takes him 10 minutes to complete. One correct way to figure out the number of hours he would need to complete this job is

- (1) $\frac{120 \text{ ft}^3}{1} \cdot \frac{10 \text{ min}}{1 \text{ load}} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{1 \text{ load}}{8 \text{ ft}^3}$
- (2) $\frac{120 \text{ ft}^3}{1} \cdot \frac{60 \text{ min}}{1 \text{ hr}} \cdot \frac{8 \text{ ft}^3}{10 \text{ min}} \cdot \frac{1}{1 \text{ load}}$
- (3) $\frac{120 \text{ ft}^3}{1} \cdot \frac{1 \text{ load}}{10 \text{ min}} \cdot \frac{8 \text{ ft}^3}{1 \text{ load}} \cdot \frac{1 \text{ hr}}{60 \text{ min}}$
- (4) $\frac{120 \text{ ft}^3}{1} \cdot \frac{1 \text{ load}}{8 \text{ ft}^3} \cdot \frac{10 \text{ min}}{1 \text{ load}} \cdot \frac{1 \text{ hr}}{60 \text{ min}}$

- 21** One characteristic of all linear functions is that they change by

- (1) equal factors over equal intervals
- (2) unequal factors over equal intervals
- (3) equal differences over equal intervals
- (4) unequal differences over equal intervals

- 22** What are the solutions to the equation $x^2 - 8x = 10$?

- (1) $4 \pm \sqrt{10}$
- (2) $4 \pm \sqrt{26}$
- (3) $-4 \pm \sqrt{10}$
- (4) $-4 \pm \sqrt{26}$

**Use this space for
computations.**

- 23** The formula for blood flow rate is given by $F = \frac{p_1 - p_2}{r}$, where F is the flow rate, p_1 the initial pressure, p_2 the final pressure, and r the resistance created by blood vessel size. Which formula can *not* be derived from the given formula?

(1) $p_1 = Fr + p_2$

(3) $r = F(p_2 - p_1)$

(2) $p_2 = p_1 - Fr$

(4) $r = \frac{p_1 - p_2}{F}$

- 24** Morgan throws a ball up into the air. The height of the ball above the ground, in feet, is modeled by the function $h(t) = -16t^2 + 24t$, where t represents the time, in seconds, since the ball was thrown. What is the appropriate domain for this situation?

(1) $0 \leq t \leq 1.5$

(3) $0 \leq h(t) \leq 1.5$

(2) $0 \leq t \leq 9$

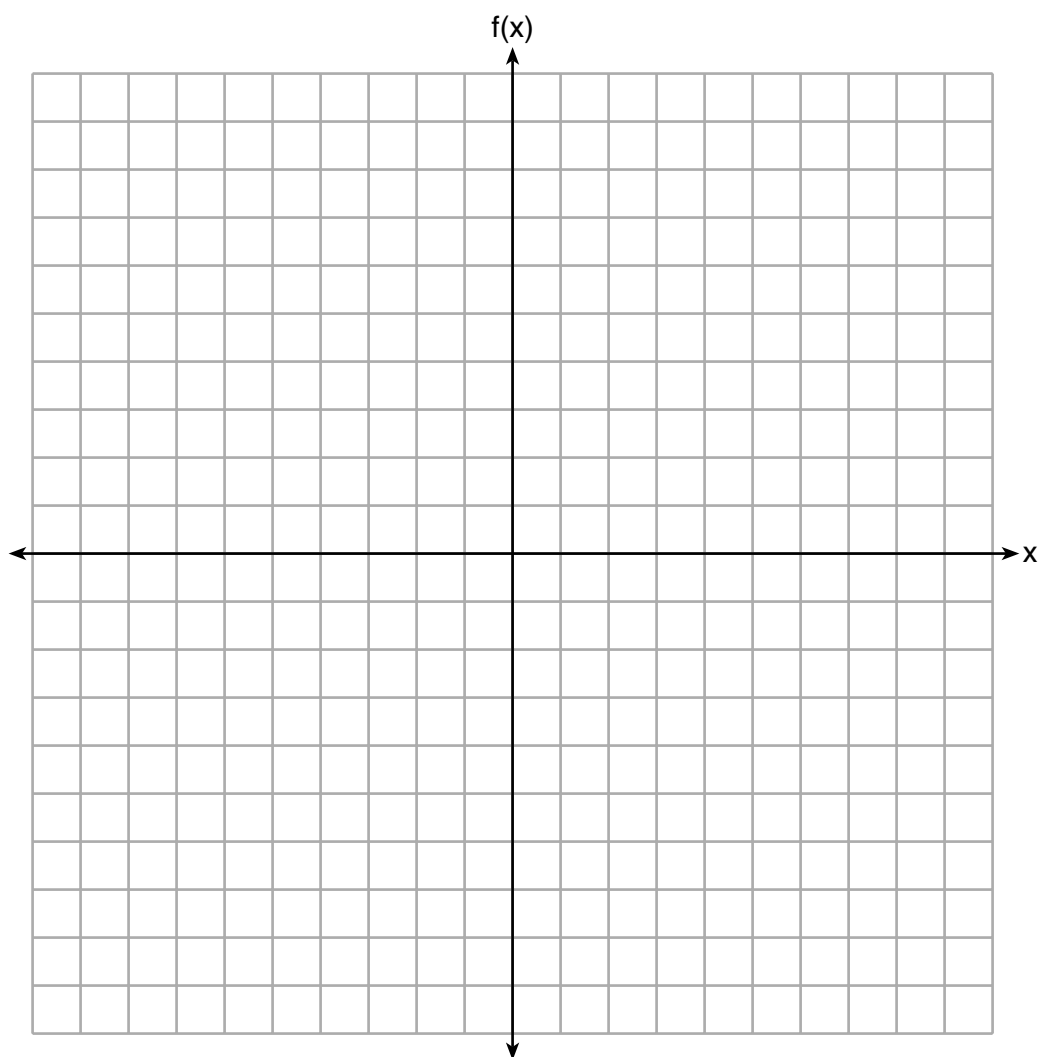
(4) $0 \leq h(t) \leq 9$

Part II

Answer all 8 questions in this part. Each correct answer will receive 2 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

25 Express in simplest form: $(3x^2 + 4x - 8) - (-2x^2 + 4x + 2)$

26 Graph the function $f(x) = -x^2 - 6x$ on the set of axes below.



State the coordinates of the vertex of the graph.

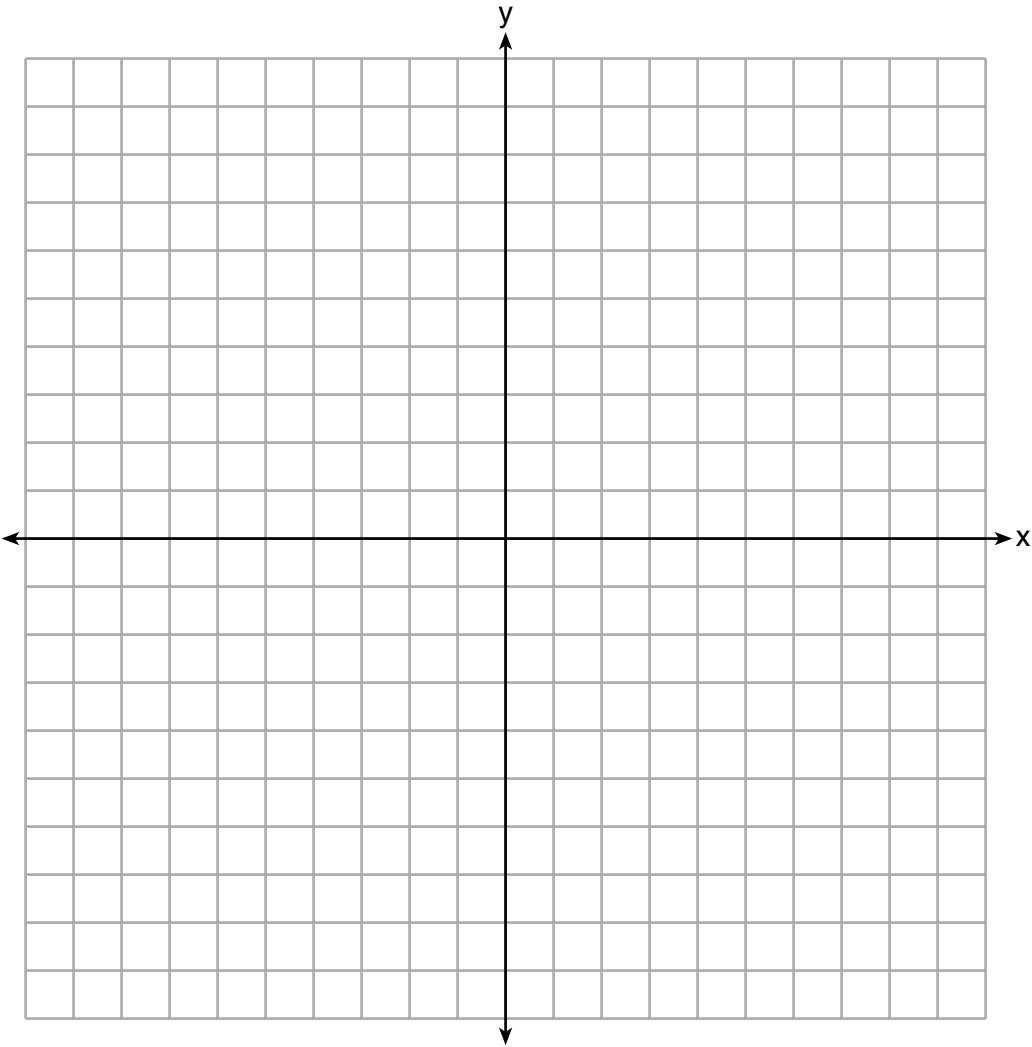
27 State whether $7 - \sqrt{2}$ is rational or irrational. Explain your answer.

28 The value, $v(t)$, of a car depreciates according to the function $v(t) = P(.85)^t$, where P is the purchase price of the car and t is the time, in years, since the car was purchased. State the percent that the value of the car *decreases* by each year. Justify your answer.

- 29** A survey of 100 students was taken. It was found that 60 students watched sports, and 34 of these students did not like pop music. Of the students who did *not* watch sports, 70% liked pop music. Complete the two-way frequency table.

	Watch Sports	Don't Watch Sports	Total
Like Pop			
Don't Like Pop			
Total			

30 Graph the inequality $y + 4 < -2(x - 4)$ on the set of axes below.



31 If $f(x) = x^2$ and $g(x) = x$, determine the value(s) of x that satisfy the equation $f(x) = g(x)$.

32 Describe the effect that each transformation below has on the function $f(x) = |x|$, where $a > 0$.

$$g(x) = |x - a|$$

$$h(x) = |x| - a$$

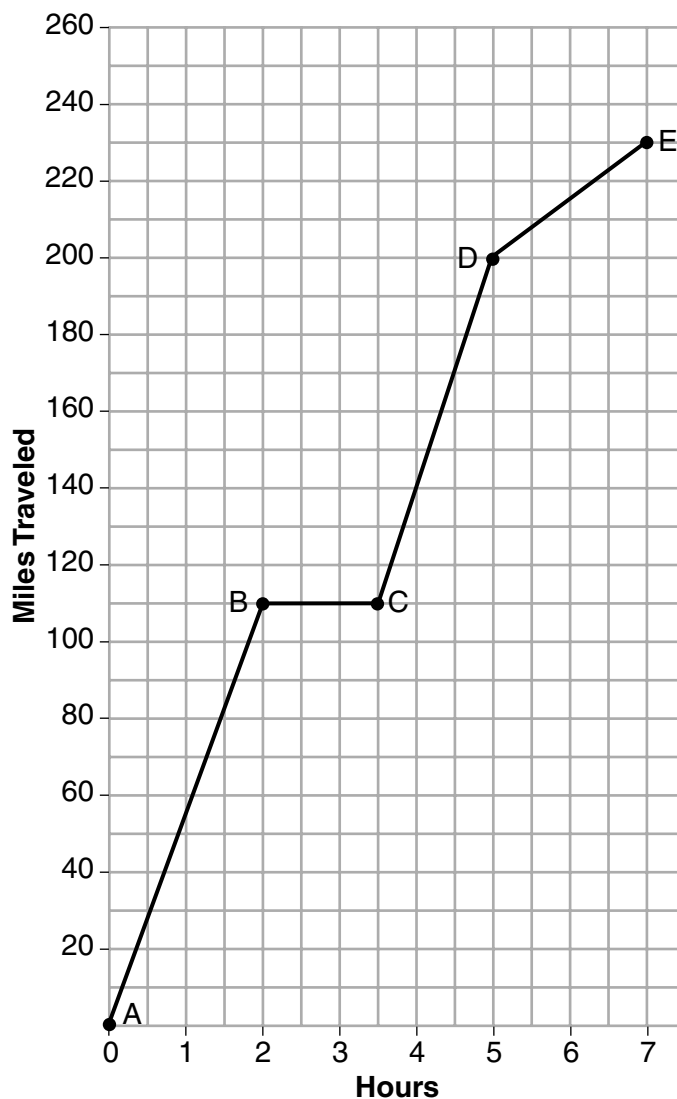
Part III

Answer all 4 questions in this part. Each correct answer will receive 4 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided for each question to determine your answer. Note that diagrams are not necessarily drawn to scale. For all questions in this part, a correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [16]

- 33 The function $r(x)$ is defined by the expression $x^2 + 3x - 18$. Use factoring to determine the zeros of $r(x)$.

Explain what the zeros represent on the graph of $r(x)$.

- 34** The graph below models Craig’s trip to visit his friend in another state. In the course of his travels, he encountered both highway and city driving.



Based on the graph, during which interval did Craig most likely drive in the city? Explain your reasoning.

Question 34 is continued on the next page.

Question 34 continued.

Explain what might have happened in the interval between B and C .

Determine Craig's average speed, to the *nearest tenth of a mile per hour*, for his entire trip.

35 Given:

$$g(x) = 2x^2 + 3x + 10$$

$$k(x) = 2x + 16$$

Solve the equation $g(x) = 2k(x)$ algebraically for x , to the *nearest tenth*.

Explain why you chose the method you used to solve this quadratic equation.

- 36** Michael has \$10 in his savings account. Option 1 will add \$100 to his account each week. Option 2 will double the amount in his account at the end of each week.

Write a function in terms of x to model each option of saving.

Michael wants to have at least \$700 in his account at the end of 7 weeks to buy a mountain bike. Determine which option(s) will enable him to reach his goal. Justify your answer.

Part IV

Answer the question in this part. A correct answer will receive 6 credits. Clearly indicate the necessary steps, including appropriate formula substitutions, diagrams, graphs, charts, etc. Utilize the information provided to determine your answer. Note that diagrams are not necessarily drawn to scale. A correct numerical answer with no work shown will receive only 1 credit. All answers should be written in pen, except for graphs and drawings, which should be done in pencil. [6]

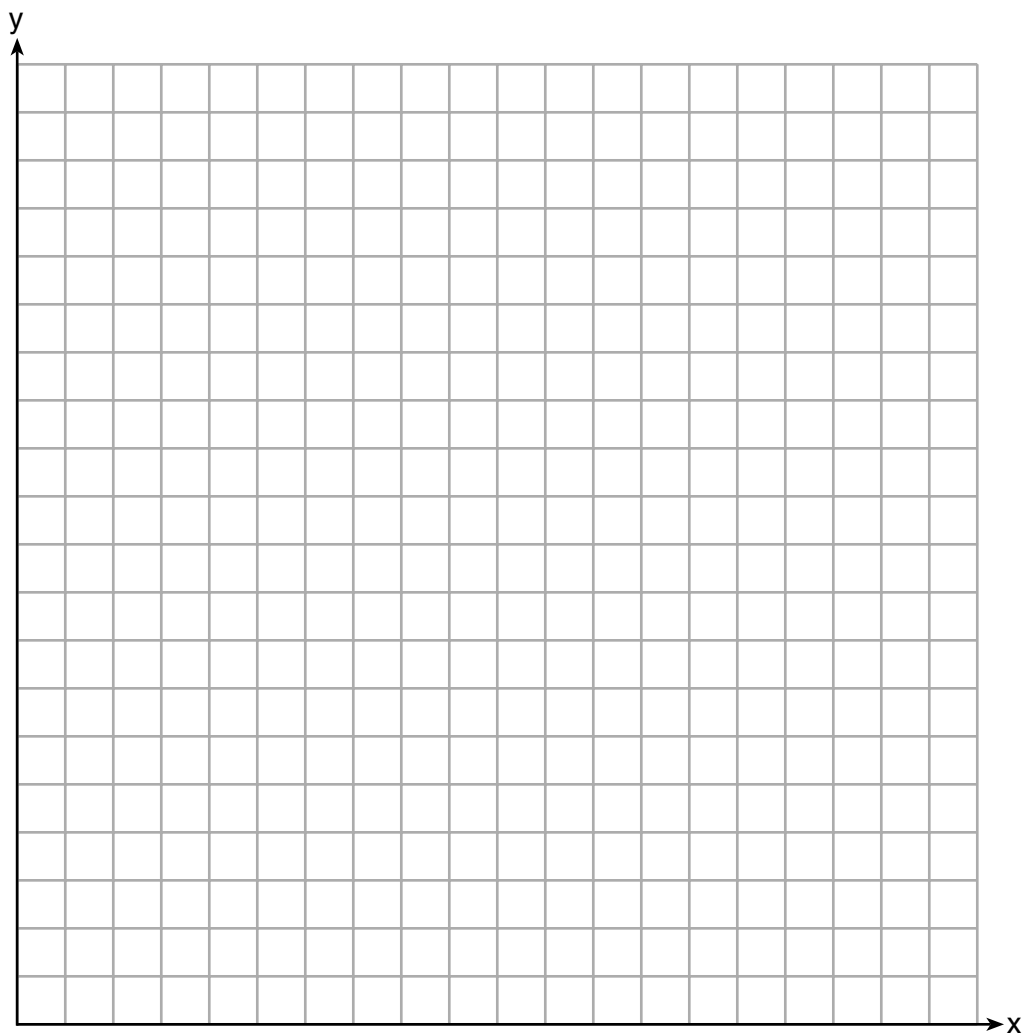
- 37** Central High School had five members on their swim team in 2010. Over the next several years, the team increased by an average of 10 members per year. The same school had 35 members in their chorus in 2010. The chorus saw an increase of 5 members per year.

Write a system of equations to model this situation, where x represents the number of years since 2010.

Question 37 is continued on the next page.

Question 37 continued.

Graph this system of equations on the set of axes below.



Explain in detail what each coordinate of the point of intersection of these equations means in the context of this problem.

If the student's responses for the multiple-choice questions are being hand scored prior to being scanned, the scorer must be careful not to make any marks on the answer sheet except to record the scores in the designated score boxes. Marks elsewhere on the answer sheet will interfere with the accuracy of the scanning.

Part I

Allow a total of 48 credits, 2 credits for each of the following.

(1) 3	(9) 3	(17) 3
(2) 2	(10) 3	(18) 2
(3) 4	(11) 1	(19) 1
(4) 2	(12) 2	(20) 4
(5) 1	(13) 4	(21) 3
(6) 3	(14) 1	(22) 2
(7) 1	(15) 4	(23) 3
(8) 1	(16) 4	(24) 1

Updated information regarding the rating of this examination may be posted on the New York State Education Department's web site during the rating period. Check this web site at: <http://www.p12.nysed.gov/assessment/> and select the link "Scoring Information" for any recently posted information regarding this examination. This site should be checked before the rating process for this examination begins and several times throughout the Regents Examination period.

The Department is providing supplemental scoring guidance, the "Model Response Set," for the Regents Examination in Algebra I (Common Core). This guidance is recommended to be part of the scorer training. Schools are encouraged to incorporate the Model Response Sets into the scorer training or to use them as additional information during scoring. While not reflective of all scenarios, the model responses selected for the Model Response Set illustrate how less common student responses to constructed-response questions may be scored. The Model Response Set will be available on the Department's web site at <http://www.nysedregents.org/algebraone/>.

**Map to the Common Core Learning Standards
Algebra I (Common Core)
June 2017**

Question	Type	Credits	Cluster
1	Multiple Choice	2	F-IF.B
2	Multiple Choice	2	A-SSE.A
3	Multiple Choice	2	A-REI.D
4	Multiple Choice	2	S-ID.C
5	Multiple Choice	2	F-IF.A
6	Multiple Choice	2	A-SSE.A
7	Multiple Choice	2	A-CED.A
8	Multiple Choice	2	A-REI.C
9	Multiple Choice	2	F-IF.A
10	Multiple Choice	2	A-APR.B
11	Multiple Choice	2	A-CED.A
12	Multiple Choice	2	F-LE.A
13	Multiple Choice	2	A-REI.B
14	Multiple Choice	2	S-ID.C
15	Multiple Choice	2	S-ID.A
16	Multiple Choice	2	F-IF.C
17	Multiple Choice	2	F-IF.C
18	Multiple Choice	2	F-IF.A
19	Multiple Choice	2	A-REI.B
20	Multiple Choice	2	N-Q.A

21	Multiple Choice	2	F-LE.A
22	Multiple Choice	2	A-REI.B
23	Multiple Choice	2	A-CED.A
24	Multiple Choice	2	F-IF.B
25	Constructed Response	2	A-APR.A
26	Constructed Response	2	F-IF.C
27	Constructed Response	2	N-RN.B
28	Constructed Response	2	F-LE.B
29	Constructed Response	2	S-ID.B
30	Constructed Response	2	A-REI.D
31	Constructed Response	2	A-REI.D
32	Constructed Response	2	F-BF.B
33	Constructed Response	4	A-SSE.B
34	Constructed Response	4	F-IF.B
35	Constructed Response	4	A-REI.A
36	Constructed Response	4	A-CED.A
37	Constructed Response	6	A-CED.A

Part II

For each question, use the specific criteria to award a maximum of 2 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(25) [2] $5x^2 - 10$ or $5(x^2 - 2)$, and appropriate work is shown.

[1] Appropriate work is shown, but one computational error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] $5x^2 - 10$, but no work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

(26) [2] A correct graph is drawn and $(-3,9)$ is stated.

[1] Appropriate work is shown, but one computational or graphing error is made.

or

[1] Appropriate work is shown, but one conceptual error is made.

or

[1] A correct graph is drawn, but no further correct work is shown.

or

[1] $(-3,9)$, but no further correct work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (27) [2] Irrational, and a correct explanation is written.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] 5.585786438 and irrational are written, but the explanation is missing or incorrect.
- [0] Irrational is written, but no explanation is written.
- or**
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (28) [2] 15, and a correct justification is written.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] 15, but the justification is missing or incorrect.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (29) [2] The frequency table is completed correctly.
- [1] Appropriate work is shown, but one computational error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- [0] Only the given information of 100, 60, and 34 is written in the table.
- or**
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (30) [2] The inequality is graphed and shaded correctly.
- [1] Appropriate work is shown, but one computational or graphing error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- [0] $y + 4 = -2(x - 4)$ is graphed correctly, but no further correct work is shown.
- or**
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
-
- (31) [2] 0 and 1, and correct work is shown.
- [1] Appropriate work is shown, but one computational or factoring error is made.
- or**
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] Appropriate work is shown to find $x(x - 1) = 0$, but no further correct work is shown.
- or**
- [1] Appropriate work is shown, but only one solution is found.
- or**
- [1] Appropriate work is shown, but the solutions are written as $(0,0)$ and $(1,1)$.
- or**
- [1] 0 and 1, but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
-
- (32) [2] $f(x)$ is shifted right by a and $f(x)$ is shifted down by a are stated.
- [1] Appropriate work is shown, but one conceptual error is made.
- or**
- [1] Only one shift is stated correctly.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
-

Part III

For each question, use the specific criteria to award a maximum of 4 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

- (33) [4] -6 and 3 , and correct work is shown, and a correct explanation is written.
- [3] Appropriate work is shown, but one computational or factoring error is made.
- or**
- [3] Appropriate work is shown, but an incomplete explanation is written.
- [2] Appropriate work is shown, but two or more computational or factoring errors are made.
- or**
- [2] Correct work is shown to find -6 and 3 , but no explanation is written.
- or**
- [2] A correct explanation is written, but no further correct work is shown.
- [1] -6 and 3 , but a method other than factoring is used and no further correct work is shown.
- or**
- [1] -6 and 3 , but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
- (34) [4] D to E with a correct explanation is written, a correct explanation for interval B to C is written, and 32.9 .
- [3] Appropriate work is shown, but one explanation is missing or incorrect.
- [2] D to E and 32.9 are stated, but no further correct work is shown.
- [1] D to E or 32.9 is stated, but no further correct work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (35) [4] 3.6 and -3.1 , and correct algebraic work is shown, and a correct explanation is written.
- [3] Appropriate work is shown, but one computational or rounding error is made.
- or**
- [3] Correct work is shown to find 3.6 and -3.1 , but the explanation is missing or incorrect.
- or**
- [3] Appropriate work is shown, but only one correct root is stated.
- or**
- [3] Appropriate work is shown, but the roots are not expressed in decimal form.
- [2] Appropriate work is shown, but two or more computational or rounding errors are made.
- or**
- [2] Appropriate work is shown to find 3.6 or -3.1 , but the explanation is missing or incorrect.
- or**
- [2] 3.6 and -3.1 are found using a method other than algebraic, but an appropriate explanation is written.
- [1] A correct substitution into the quadratic formula is made, but no further correct work is shown.
- or**
- [1] 3.6 and -3.1 , but no work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

- (36) [4] $f(x) = 10 + 100x$ and $g(x) = 10(2)^x$ or equivalent functions, “both” is stated, and a correct justification is given.
- [3] Appropriate work is shown, but one computational error is made.
- or***
- [3] Appropriate work is shown, “both” is stated, but the justification is missing or incorrect.
- or***
- [3] Appropriate work is shown, but “both” is not stated.
- or***
- [3] Appropriate work is shown, but two expressions are written instead of equations.
- [2] Appropriate work is shown, but two or more computational errors are made.
- or***
- [2] Correct functions are stated, but no further correct work is shown.
- or***
- [2] “Both” is stated and a correct justification is given, but no further correct work is shown.
- [1] “Both” is stated, but no work is shown.
- or***
- [1] One correct function is written, but no further correct work is shown.
- [0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.

Part IV

For this question, use the specific criteria to award a maximum of 6 credits. Unless otherwise specified, mathematically correct alternative solutions should be awarded appropriate credit.

(37) [6] $y = 10x + 5$ and $y = 5x + 35$ are written and graphed correctly, and at least one is labeled, and a correct explanation is written.

[5] Appropriate work is shown, but one graphing or labeling error is made.

or

[5] Appropriate work is shown, but the explanation for one of the coordinates is missing or incorrect.

or

[5] Appropriate work is shown, but expressions are written instead of equations.

[4] Appropriate work is shown, but the explanation for each coordinate is missing or incorrect.

[3] Appropriate work is shown, but one graphing or labeling error is made and the explanation is missing or incorrect.

[2] A correct system of equations is written, but no further correct work is shown.

or

[2] A correct explanation for both coordinates is written, but no further correct work is shown.

or

[2] One equation is written and graphed correctly, but no further correct work is shown.

[1] One correct equation is written, but no further correct work is shown.

or

[1] (6,65) is stated, but no further correct work is shown.

[0] A zero response is completely incorrect, irrelevant, or incoherent or is a correct response that was obtained by an obviously incorrect procedure.
